

EventSync

Event Registration & Venue Scheduling System

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A database-connected desktop system
for reliable event operations.

EVENTS	SESSIONS	VENUES
PARTICIPANTS	REGISTRATIONS	
ATTENDANCE	PAYMENTS	
CERTIFICATES	REPORTS	

Project Recap & Objectives

THE PROBLEM

Institutions need one reliable system to coordinate event registration, venues, participants, attendance, payments and certificates.



OBJECTIVES

- Centralize event and participant information
- Manage sessions and venue allocation
- Prevent duplicate registrations
- Track attendance and payments
- Determine certificate eligibility
- Generate meaningful reports
- Provide a simple database-connected application

System Architecture & Technology Stack



Desktop UI → application logic → connector → relational database

TECHNOLOGY STACK

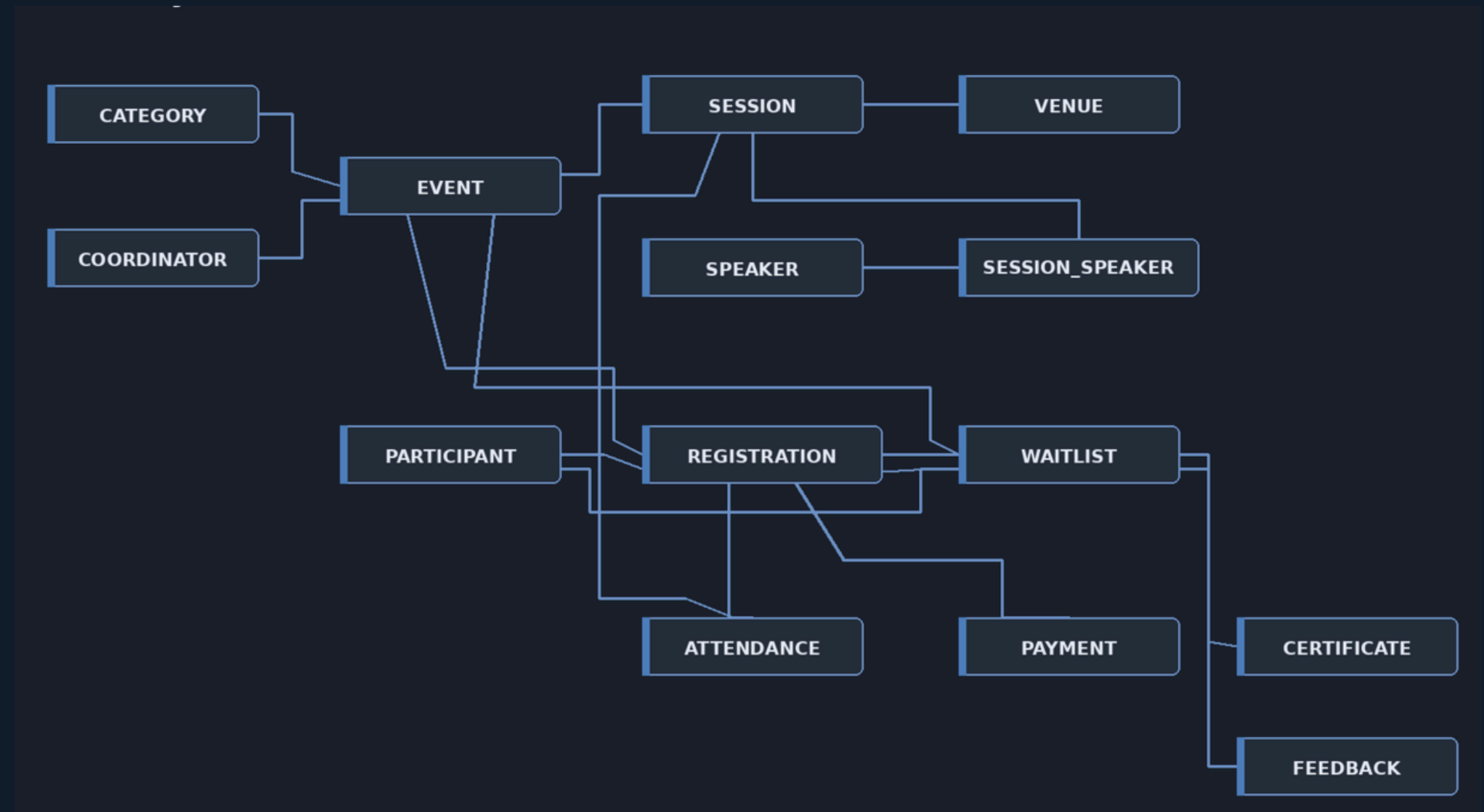


Database Design

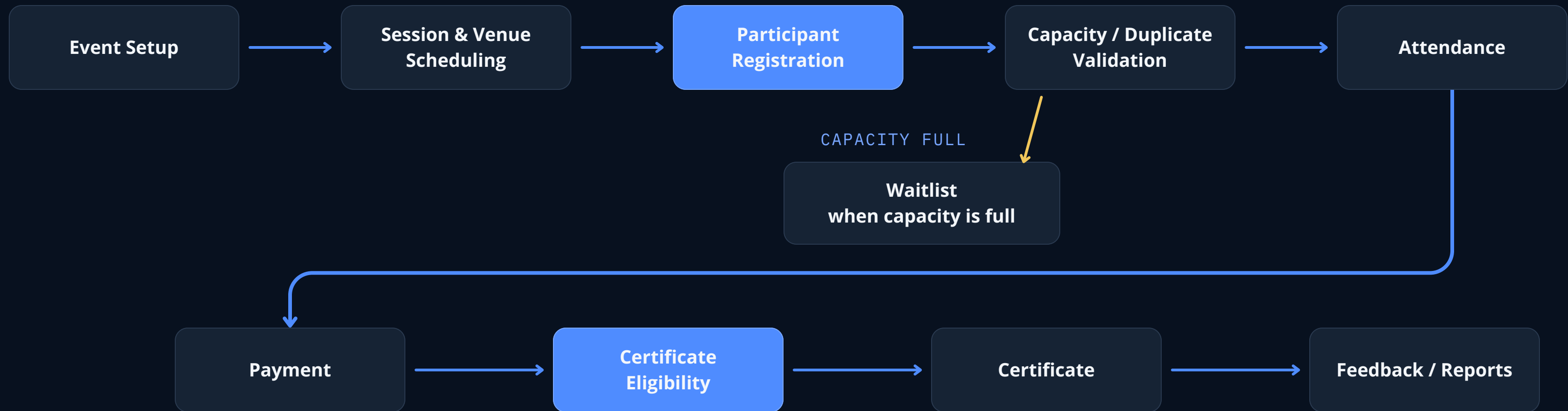
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RELATIONAL TABLES

- Primary & foreign keys
- UNIQUE • NOT NULL • CHECK
- Events ↔ sessions ↔ venues
- Participants ↔ registrations
- Normalized to Third Normal Form



Application Workflow



Major Features

Event Management Create, edit, search and delete events	Participant Management Maintain complete participant records	Session Scheduling Validate event-linked session timing	Venue Allocation Capacity and overlap controls	Registration & Waitlist Prevent duplicates; preserve queue
Attendance Record participation by registration	Payments Validate and track non-negative amounts	Certificates Eligibility-aware generation	Search & Reporting Fast retrieval and useful views	Validation & Errors Clear database and UI safeguards

Event & Session Management

- Add, edit, search and delete events
- Link sessions to events and venues
- Validate session start and end times
- Check venue capacity against attendance
- Detect venue overlap with time-overlap logic
- Allow back-to-back sessions when no overlap exists

ES

EventSync

Event Management System

Dashboard

Events

Sessions

Participants

Registration

Attendance

Payments

Certificates

Events

Create and maintain event details

Event Details

Event Name:

Category:

Coordinator:

Date (YYYY-MM-DD):

Status:

Open

Max Capacity:

Registration Fee:

Description:

ACTIONS

Add

Update

Delete

Clear

Search by Name:

Search

Show All

ID	Name	Category	Coord.	Date	Cap.	Fee	Status	Desc
3	Cultural Fest 2026	Cultural	Priya Nair	2026-12-0	500	0.00	Open	Annual ui
2	Future Tech Semir	Seminar	Rahul Mehta	2026-11-1	200	100.00	Open	Seminar
1	AI Innovation Wor	Workshop	Ananya Sharm	2026-11-1	100	200.00	Open	Hands-on

Events Tab

ES

EventSync

Event Management System

Dashboard

Events

Sessions

Participants

Registration

Attendance

Payments

Certificates

Sessions

Schedule sessions and venues

Session Details

Session Title:

Event:

Venue:

Date (YYYY-MM-DD):

Capacity:

-

Start Time:

End Time:

Expected Attendance:

ACTIONS

Add

Update

Delete

Clear

ID	Title	Event	Venue	Date	Start	End	Expected
4	Cultural Performa	Cultural Fest 2026	Open Amphitheat	2026-12-05	18:00:00	21:00:00	400
3	Emerging Technol	Future Tech Semi	Seminar Hall 1	2026-11-15	10:00:00	12:00:00	120
1	Introduction to AI	AI Innovation Wo	Innovation Lab	2026-11-10	10:00:00	11:30:00	80
2	Machine Learning	AI Innovation Wo	Innovation Lab	2026-11-10	12:00:00	14:00:00	80

Sessions Tab

Registration, Waitlist, Attendance & Payments



Duplicates blocked • waitlist positions maintained • attendance recorded per registration/session • payments validated ≥ 0

Registration

Manage registrations and the waitlist

Registration Desk

REGISTRATION DETAILS

Event:

Participant:

Date (YYYY-MM-DD):

2026-10-04

Status:

Confirmed

Event Capacity Info

Maximum Capacity: -

Registered: -

Seats Remaining: -

ACTIONS

Register

Clear Form

Delete Registration

Delete Waitlist

Search (Event or Participant):

Search

Show All

Registrations

ID	Event	Participant	Date	Status
6	Cultural Fest 2026	Kabir Joshi	2026-11-01	Confirmed
5	Future Tech Seminar	Ishita Rao	2026-10-23	Confirmed
4	Future Tech Seminar	Aarav Patel	2026-10-22	Confirmed
2	AI Innovation Workshop	Meera Shah	2026-10-21	Confirmed

Waitlist

Wait ID	Event	Participant	Date	Position	Status
1	AI Innovation Workshop	Ishita Rao	2026-10-25	1	Waiting
2	AI Innovation Workshop	Kabir Joshi	2026-10-26	2	Waiting

Registration Tab

Attendance

Record and review session attendance

Mark Attendance

Event:

Session:

Reg	Participant	Status	Check-in
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Mark Present

Mark Absent

Attendance Records

Filter:

Search

Show All

Att	Event	Session	Participant	Status	Time
14	Future Tech Seminar	Emerging Technologies	Aarav Patel	Present	9:50:00
13	AI Innovation Workshop	Machine Learning Works	Rohan Gupta	Present	12:01:00
12	AI Innovation Workshop	Introduction to AI	Rohan Gupta	Present	9:58:00
11	AI Innovation Workshop	Machine Learning Works	Meera Shah	Absent	-
10	AI Innovation Workshop	Introduction to AI	Meera Shah	Present	10:02:00
9	AI Innovation Workshop	Machine Learning Works	Aarav Patel	Present	11:55:00
8	AI Innovation Workshop	Introduction to AI	Aarav Patel	Present	9:55:00

Delete Selected Record

Attendance Tab

Payments

Track registration payments

Payment Details

Registration:

Amount: Payment Method:

UPI

Payment Date:

2026-10-04

 Payment Status:

Paid

Payment Summary

Total Payments: 6

Total Revenue: Rs.800.00

ACTIONS

Add Payment

Update

Delete

Clear Form

Search (Event or Participant):

Search

Show All

Pay	Reg	Event	Participant	Amount	Date	Method	Status
6	6	Cultural Fest 2026	Kabir Joshi	0.00	2026-11-01	Free	Paid
5	5	Future Tech Seminar	Ishita Rao	100.00	2026-10-23	Card	Paid
4	4	Future Tech Seminar	Aarav Patel	100.00	2026-10-22	UPI	Paid
3	3	AI Innovation Worksho	Rohan Gupta	200.00	2026-10-21	UPI	Paid
2	2	AI Innovation Worksho	Meera Shah	200.00	2026-10-21	Card	Paid
1	1	AI Innovation Worksho	Aarav Patel	200.00	2026-10-20	UPI	Paid

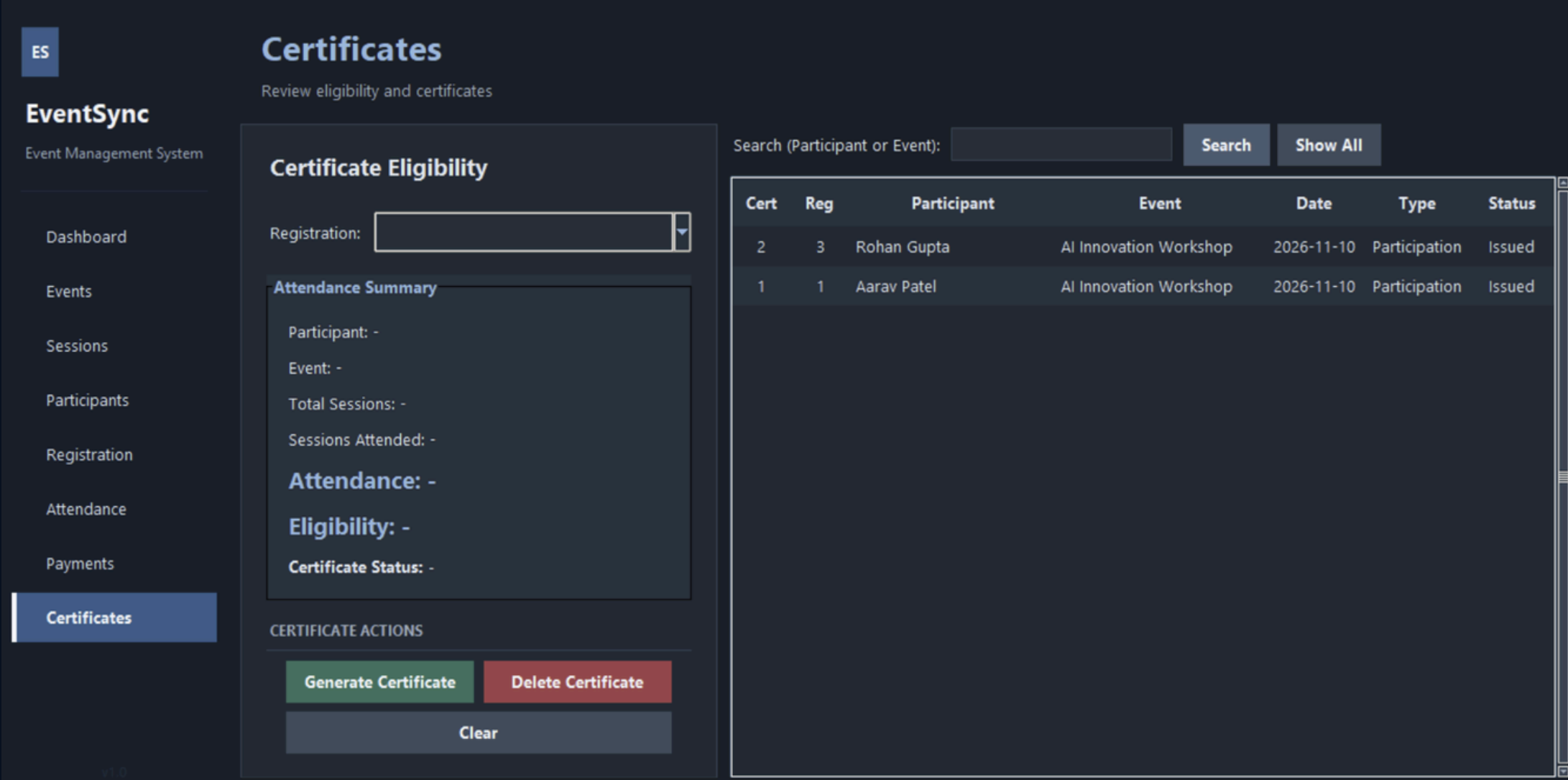
Payments Tab

Certificate Eligibility & Reporting

ATTENDANCE ≥ 75%

ELIGIBLE

- Calculated from attendance records
- Ineligible participants cannot generate certificates
- Duplicate certificates are prevented
- Certificate status is stored in the database
- Reports and views available where appropriate



Certificates Tab

Testing & Validation

Duplicate registration	BLOCKED	Venue capacity violation	DETECTED
Negative payment	BLOCKED	Venue overlap	DETECTED
Invalid feedback rating	BLOCKED	Certificate eligibility	VERIFIED
Invalid session time	BLOCKED	Duplicate certificate	BLOCKED
Invalid foreign key	BLOCKED	Search functionality	VERIFIED

Final database restored to original sample state after testing.

Conclusion & Future Scope

CONCLUSION

- Database-backed event management system implemented
- 3NF relational design integrated with a Python desktop app
- CRUD, validation, scheduling and participant lifecycle delivered
- Systematically tested across business and database rules

FUTURE SCOPE

- Automated email notifications
- Online registration and payment
- Role-based authentication
- Automatic waitlist promotion
- Advanced analytics and dashboard reports
- Web and mobile version

Thank You

Questions?